FRONT

Curve $v=\sqrt{gr\frac{\tan\theta\pm\mu_s}{1\mp\mu_s\tan\theta}}$ (+ max, − min).
Flat: $v_{max}=\sqrt{\mu_s gr}$. No friction: $v=\sqrt{gr\tan\theta}$.

Loop top barely (N=0): $v_t=\sqrt{gR}$; bottom min $v_b=\sqrt{5gR}$; launch $h=\frac{5}{2}R$.
$v_t^2=v_b^2-4gR=2g(h-2R)$. $N_{top}=\frac{mv_t^2}{R}-mg$, $N_{bot}=\frac{mv_b^2}{R}+mg$.

Energy $E_i+E_{add}=E_f+E_{lost}$; $E_{lost}=\mu Nd$; incline $N=mg\cos\theta$. Never through a collision.

Fall / coaster (no friction)
$v_f=\sqrt{v_i^2+2g(h_i-h_f)}$.

Halfpipe $h_f=h_i+\frac{v_i^2}{2g}-n\mu L$ (n = crossings of rough length L). Full crossings $=\lfloor h_i/\mu L\rfloor$.

One block on incline (Δx along slope)
Up: $v_f^2=v_i^2-2g\Delta x(\sin\theta+\mu\cos\theta)$
Down: $v_f^2=v_i^2+2g\Delta x(\sin\theta-\mu\cos\theta)$
Stop up: $\Delta x=\frac{v_i^2}{2g(\sin\theta+\mu\cos\theta)}$
Stop down: $\Delta x=\frac{v_i^2}{2g(\mu\cos\theta-\sin\theta)}$
$\sqrt{-}$ = stops first. Time: $t=\frac{2\Delta x}{v_i+v_f}$.

Spring $\frac{1}{2}kx^2=\frac{1}{2}mv^2+mgh$. Stop a mass: $x=\sqrt{\frac{2mgh+mv_i^2}{k}}$; launch up $h=\frac{kx^2}{2mg}$ (from release pt); level $v=x\sqrt{k/m}$.

Power $P=\frac{W}{t}=\frac{mgh}{t}=Fv$. Work $W=Fd\cos\phi$.

Atwood / mass on pulley (m_1 falls, m_2 rises; one mass: m_2=0) $v_f^2=v_i^2+\frac{2gh(m_1-m_2)}{m_1+m_2+I/R^2}$, $a=\frac{(m_1-m_2)g}{m_1+m_2+I/R^2}$, $T=m_1(g-a)$. I/R^2: massless 0, disk $\frac{1}{2}M$, ring M, given $kMR^2\to kM$.

Incline + hanging mass + pulley
$v_f^2=v_i^2+\frac{2g\Delta xB}{m_1+m_2+I/R^2}$ (m_1 on slope, m_2 hangs).
m_2 falls: $B=m_2-m_1\sin\theta-\mu m_1\cos\theta$
m_1 slides down: $B=m_1\sin\theta-m_2-\mu m_1\cos\theta$
Stops: $\Delta x=\frac{-(m_1+m_2+I/R^2)v_i^2}{2gB}$.

Projectile $H=h_0+\frac{(v_0\sin\theta)^2}{2g}$, $t_{top}=\frac{v_0\sin\theta}{g}$.

Cut along the box. Print at 100% / Actual size.

BACK

Force from Δp $F=\frac{m(v_f-v_i)}{t}$. Bounce back: $\frac{2mv}{t}$; stop: $\frac{mv}{t}$.

Stick (inelastic) $v_f=\frac{m_1v_1+m_2v_2}{m_1+m_2}$; $\frac{K_f}{K_i}=\frac{(m_1+m_2)v_f^2}{m_1v_1^2+m_2v_2^2}$ (one at rest: $\frac{m_1}{m_1+m_2}$).

Elastic $v_{1f}=\frac{(m_1-m_2)v_{1i}+2m_2v_{2i}}{m_1+m_2}$, $v_{2f}=v_{1i}+v_{1f}-v_{2i}$. Equal m, 2 at rest: $v_{1f}=0$, $v_{2f}=v_{1i}$.

Ballistic $v=\frac{M+m}{m}\sqrt{2gh}$; $V=\frac{mv}{M+m}$, $h=\frac{V^2}{2g}$; $\frac{K_f}{K_i}=\frac{m}{M+m}$.

Rot. kinematics $\omega_f=\omega_i+\alpha t$; $\Delta\theta=\frac{\omega_i+\omega_f}{2}t$; $\omega_f^2=\omega_i^2+2\alpha\Delta\theta$; $\Delta\theta=\omega_it+\frac{1}{2}\alpha t^2$. Use the one without the quantity you lack.

***I* (center)** ring MR^2, disk $\frac{1}{2}MR^2$, sphere $\frac{2}{5}MR^2$, rod mid $\frac{1}{12}m\ell^2$, rod end $\frac{1}{3}m\ell^2$, point mr^2.

Parallel axis $I=I_{cm}+md^2$, d from CENTER. Rod, axis a from end: $I=m[\frac{\ell^2}{12}+(\frac{\ell}{2}-a)^2]$. Disk about rim $\frac{3}{2}MR^2$. Pieces add.

Torque $v=R\omega$, $a=R\alpha$, $s=R\theta$. $\tau=rF\sin\phi=I\alpha$. Pulley: $RT=I\alpha\Rightarrow T=\frac{I}{R^2}a$. $K_{rot}=\frac{1}{2}I\omega^2$.

Rolling $v=\sqrt{\frac{2gh}{1+I/mR^2}}$, $\omega=\frac{v}{R}$, $a=\frac{g\sin\theta}{1+I/mR^2}$. Ring $\sqrt{gh}$, disk $\sqrt{\frac{4}{3}gh}$, sphere $\sqrt{\frac{10}{7}gh}$, slide $\sqrt{2gh}$.

Angular momentum $(L_i=L_f)$ $\omega_f=\frac{I_1\omega_1+I_2\omega_2\pm mvr}{I_1+I_2+I_{add}}$. $\omega_2<0$ counter-rotating; $-mvr$ runs against rim; drop/step on: no mvr. I_{add}: person mr^2, rod center→edge $\frac{1}{3}m\ell^2$, disk $\frac{1}{2}mr^2$. Walk in/out: $\omega_f=\frac{I_i}{I_f}\omega_i$; rim→center $\omega_f=\frac{M+2m}{M}\omega_i$. $K=\frac{1}{2}I\omega^2$.

Pick the law speed/height: energy; circle: $\Sigma F=\frac{mv^2}{r}$; hit: momentum; time: forces; spin change: $L_i=L_f$.

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